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Analysis and prediction of backfat thickness in gestating sows using machine learning algorithms

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ABSTRACT

Sow backfat (BF) thickness is a key indicator for predicting the nutrient requirements and influencing on reproductive performance of gestating sows. The purpose of this study was to determine feature importance for healthy piglets, define the optimal BF at farrowing and change trend of BF during gestation, as well as to establish the prediction models of BF in gestating sows using 10 machine learning (ML) models. A database with 64,298 observations including 3 categorical and 18 numerical features was used for data analysis and modeling. Compared to other features, BF at farrowing was the most important feature for healthy piglets. The optimal BF at farrowing of parity 1, 2, and ≥3 was 18 mm, 16 mm, and 16 mm, respectively, and the early to middle stage of gestation was the best period for body condition restoration. The eXtreme gradient boosting regression (XGBR) and gradient boosting regression (GBR) exhibited best prediction performance with lowest RMSE (30d, 60d, 90d and farrowing of gestation were 1.17, 1.09, 1.01 and 0.81 mm, respectively) and MAPE (30d, 60d, 90d and farrowing of gestation were 5.57 %, 4.93 %, 4.44 % and 3.54 %, respectively), showing best accuracy and stability among the 10 ML models. The analysis and prediction of BF during gestation based on ML methods provide technical support for accurately predicting nutrient requirements and formulating precise feeding strategy of gestating sows.

1. Introduction

The reproductive performance of sows is one of the most critical factors in the swine industry, closely related to the production profitability of breeding farms. The nutritional status, parity, environment, breed, and body condition of sows stands out as pivotal roles in influencing reproductive performance [1,2]. Particularly, backfat (BF) serves as a vital indicator of nutritional reserve and body condition of sows, and has been identified as a key factor in optimizing the reproductive performance, such as the birth and weaning weight of piglets, vitality of neonatal piglets and the longevity of sows [2–6]. The excessive loss of BF during the lactation can decrease growth rate of piglets and total born in next cycle, as well as prolong weaning-to-estrus interval [2,7,8]. Excessive BF during gestation could generate placental inflammation, reducing total born, birth weight of piglets, feed intake and weaning performance of sows during lactation [6,9,10]. Furthermore, excessive backfat at farrowing (BFf) can cause dystocia in sows, increasing the stillborn piglets and the culling rate of sows [11,12]. Therefore, maintaining appropriate BF is necessary for optimal reproductive performance of sows [6,13]. In addition, BF is an important parameter to estimate body protein and fat during gestation for accurately predicting nutrient requirements by targeting BF during gestation, providing an important basis for formulating and adjusting the feeding strategies [14,15].

BF is considered to reflect the nutritional and metabolic status of sows before parturition. It is crucial to attain the optimal BFf to ensure

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sufficient energy reserves to support parturition and lactation, showing more significant impact on reproductive performance for sows compared to other gestation periods [2,16]. However, the recommended ideal BFf varies dramatically in different studies and institutions. In past studies, the optimal BFf was typically in the range of 17∼22 mm [6,13,17–20]. However, sows presented more lean and less fat deposition with the progress of genetic selection in recent years. SEGES Innovation had down-regulated the ideal BFf from 16∼19 mm to 14∼17 mm by analysis the relationship between BFf and performance [21]. The differences of recommended BFf probably attributed to genotype, but whether the optimal BFf was consistent in different parities had not been explored. In addition, the feeding program in production is formulated by measuring the BF at different gestation periods and dividing into different categories, aiming to achieve a higher proportion of sows attain the optimal BFf range before parturition in commercial breeding farms [22].

Visual inspection and ultrasound detection are commonly used for BF determination [23]. Visual inspection can quickly assess the body condition of sows, but has strong subjectivity and low accuracy [24]. Compared with the visual method, ultrasound device measurement of sow BF has significantly improved accuracy. However, ultrasound detection is time-consuming and labor-intensive, and sows are frequently contacted which bring to biosecurity risks [22]. Recently, visual assessment of images and the application of related deep learning methods are employed to evaluate the behavior, body weight and body size of pigs, made it possible to establish non-contact, real-time and intelligent methods to measure BF by computer vision [22,24–26]. However, there are still many limitations in applying computer vision to predict BF. Firstly, the database is relatively small in related studies and the accuracy of the model need to be improved. Furthermore, the installation of cameras and image acquisition are limited by sow activity, ceiling height and light intensity in commercial breeding farms, resulting in the limited in image acquisition and decrease in the prediction accuracy [27].

Previous studies have established regression models for predicting BF of sows through body condition scoring and machine vision methods [22,23].Young et al. [28] simultaneously measured the BF and body condition scores of 731 sows in commercial breeding farms and established a linear regression model between sow body condition scores and BF. However, body condition score was poorly correlated (R^2=0.19) with BF, indicating that body condition scoring display relatively low accuracy for BF prediction. Jian et al. [29] used image recognition to establish a linear regression model for BF estimation based on the area of the fitted ellipse (AFE), the pixel area of the sow's buttocks (AB), the enclosing rectangle pixel width (RW) and the fitted ellipse major axis pixel length (EMA), and showed good performance of R^2 and MAPE were 0.89 and 5.73 %. Yu et al. [30] and Li et al. [24] used computer vision to construct Convolutional Neural Network (CNN) models to estimate sow BF with the R2, RMSE and MAPE were 0.72 and 0.74, 1.50 mm and 1.05 mm, 7.57 % and 4.87 % respectively. Although computer vision methods presented good predictive performance, the application of computer vision in commercial breeding farms was restricted due to the limited datasets and constraints of pigsty conditions. Moreover, the above-mentioned methods primarily evaluate the real-time BF of sows but unable to predict BF and its changes in other gestation periods, leading to those methods can't be utilized to assess the nutrients requirement for gestating sows or adjust the feeding strategy in advance. Therefore, more accurate and convenient methods are essential to evaluate BF during gestation. Several factors such as genotype, environment, nutrition and feeding management have influenced on BF, and considering the limitations of conventional linear regression models in dealing with complex factors, machine learning (ML) algorithms can accurately and effectively handle the nonlinear relationship between predictor variables and response variables, significantly improving the prediction accuracy [31,32]. Therefore, establishing BF prediction models based on machine learning methods is of great significance for accurately predicting nutrient requirements of sows during gestation, implementing and formulating precise feeding during gestation, and maximizing the reproductive performance and economic benefits of sows.

The purpose of this study is to analyze the feature importance ranking for healthy piglets (HP) based on BF, reproductive performance and other information with a database >60,000. Moreover, we aim to determine the optimal BFf in different parities and explicit changing regular of BF during gestation, and evaluate the prediction performance of BF under 10 machine learning models to establish a rapid and accurate prediction model for gestating sows.

2. Methods and materials

2.1. Data collection and preparation

A total of 78,733 complete records from 26 breeding farms of Guangxi State Farms Yongxin Livestock Husbandry Group Co., Ltd. from January 2022 to January 2023 were collected. The records included basic information, BF during the gestation, feed and nutrients intake and litter performance of sows. Cleaning steps of the database have been implemented to remove unreliable and uncommon observations. Observations presenting abnormal BF with lower than 7 mm and higher than 30 mm, as well as uncommon feed intake lower than 1.5 kg and higher than 4.5 kg were removed. The final data set contained 64,298 complete observations.

2.2. Data preprocessing and analysis

The dataset consisted of 3 categorical features, namely breed, parity, and mating season (MS) and 18 numerical features, namely mating age (MA), BF at insemination, 30d, 60d, 90d and farrowing (BFi, BF30, BF60, BF90, BFf), average daily metabolizable energy (ME) and standardized ileal digestible lysine (SID-Lys) during gestation, total born of piglets per litter (TB) and HP.

To clarify the influence and importance of different factors on the HP of sow reproductive performance, permutation importance and SHapley Additive exPlanations (SHAP) were used for feature importance ranking. Permutation importance is a method that calculates the importance score of each feature variable of the dataset. This study permutation importance was performed by using XGboost model, the included features were ranked based on their information gain to clarify the influence of different features on HP. The SHAP approach display high potential for rationalizing the predictions made by complex ML models. In the prediction process of each sample using SHAP method, we can observe the influence of each feature on the prediction result.

The relationship between HP of different parities and BFf was evaluated by regression analysis to determine the optimal BFf based on the feature importance ranking. Furthermore, we used boxplots of BF trend change to illustrate the recovery and growth patterns of body condition of sows during gestation.

In order to reduce the impact of noise and outliers of database, we standardized the data to improve the stability of the model. The definition of Z-score as follows (Eq. (1)):

$$Z = \frac{x - \mu}{\sigma} \tag{1}$$

Z, x, μ, σ represent the standardized value, value to normalize, mean and standard deviation, respectively.

2.3. Selection of machine learning algorithms and optimization of the hyperparameters

Predicting BF involved to regression of continuous values. Three categorical input variables include breed, parity and MS were ordinal encoded prior to training, and then training the model combine with continuous input variables such as MA, BF, ME, SID-Lys and TB. Ten ML

Table 1
Hyperparameters selection and data transformations.

Models	Preprocessing of features	Hyperparameters name	Hyperparameters values
Linear Regression	Standardization	fit_intercept	True
Bayesian Ridge	Standardization	fit_intercept	True
		n_inter	300
		tol	0.001
Random Forest Regression	Standardization	max_depth	10
		min_samples_leaf	4
		n_estimators	200
		criterion	absolute_error
		min_samples_split	6
ElasticNet	Standardization	alpha	0.5
		l1_ratio	0.5
		max_iter	500
		tol	0.001
Decision Tree Regression	Standardization	max_depth	7
		splitter	best
		min_samples_split	20
		criterion	absolute_error
Gradient Boosting Regression	Standardization	n_estimators	190
		max_depth	4
		learning_rate	0.2
XGB Regression	Standardization	learning_rate	0.08
		n_estimators	170
		max_depth	7
		subsample	0.75
		colsample_bytree	1
Bagging Regression	Standardization	base_estimator	Decision Tree Regressor
		max_samples	0.7
		n_estimators	200
K Nearest Neighbors Regression	Standardization	n_neighbors	9
		algorithm	auto
		weights	distance
Support vector Regression	Standardization	C	10
		gamma	0.1
		epsilon	0.1
		tol	0.01

algorithms were employed to predict BF at 30d, 60d, 90d and 110d respectively with Python 3.7.0 using the scikit-learn library due to the high performance in regression tasks as well as their representability of the diversity: Linear Regression (LR), Bayesian Ridge (BRI), Random Forest Regression (RFR), Elastic Net (EN), Decision Tree Regression (DTR), Gradient Boosting Regression (GBR), eXtreme Gradient Boosting Regression (XGBR), Bagging Regression (BR), K Nearest Neighbor Regression (KNN) and Support Vector Regression (SVR).

Considering that combining meta-heuristic algorithms with ML models could increase the complexity and computational load, which

Table 2
Descriptive statistical analysis in the entire sow data set[a].

Variables	Mean	SD
Number of observations	64,298	
Parity	3.33	1.50
MA, d	637.96	233.02
BF, mm		
BFi	16.33	3.48
BF30	17.04	3.06
BF60	17.66	2.71
BF90	17.65	2.55
BFf	17.86	2.59
Average daily ME intake, kcal/d		
ME_{0-30}	8039.60	2269.79
ME_{31-60}	7280.47	2080.22
ME_{61-90}	6280.20	1161.18
ME_{91-110}	9050.28	414.03
ME	7270.95	1283.92
Average daily SID-Lys intake, kcal/d		
$SID\text{-}Lys_{0-30}$	16.33	4.64
$SID\text{-}Lys_{31-60}$	14.79	4.26
$SID\text{-}Lys_{61-90}$	12.76	2.38
$SID\text{-}Lys_{91-110}$	18.38	0.95
SID-Lys during entire gestation	16.30	2.88
Litter performance		
TB	12.99	2.81
HP	11.57	2.36

[a] MA: mating age, BFi: backfat thickness at insemination, BFf: backfat thickness at farrowing, BF30: BF at 30 days of gestation, BF60: BF at 60 days of gestation, BF90: BF at 90 days of gestation, ME:average metabolizable energy intake during entire gestation, ME_{0-30}: average metabolizable energy intake from insemination to 30d of gestation, ME_{31-60}: average metabolizable energy intake from 31 to 60d of gestation, ME_{61-90}: average metabolizable energy intake from 61 to 90d of gestation, ME_{91-110}: average metabolizable energy intake from 91 to 110d of gestation, SID-Lys: average standardized ileal digestible lysine intake during entire gestation, $SID\text{-}Lys_{0-30}$: average SID-Lys intake from insemination to 30d of gestation, $SID\text{-}Lys_{31-60}$: average SID-Lys intake from 31 to 60d of gestation, $SID\text{-}Lys_{61-90}$: average SID-Lys intake from 61 to 90d of gestation, $SID\text{-}Lys_{91-110}$: average SID-Lys intake from 91 to 110d of gestation, TB: total born of piglets per litter; HP: heathy piglets.

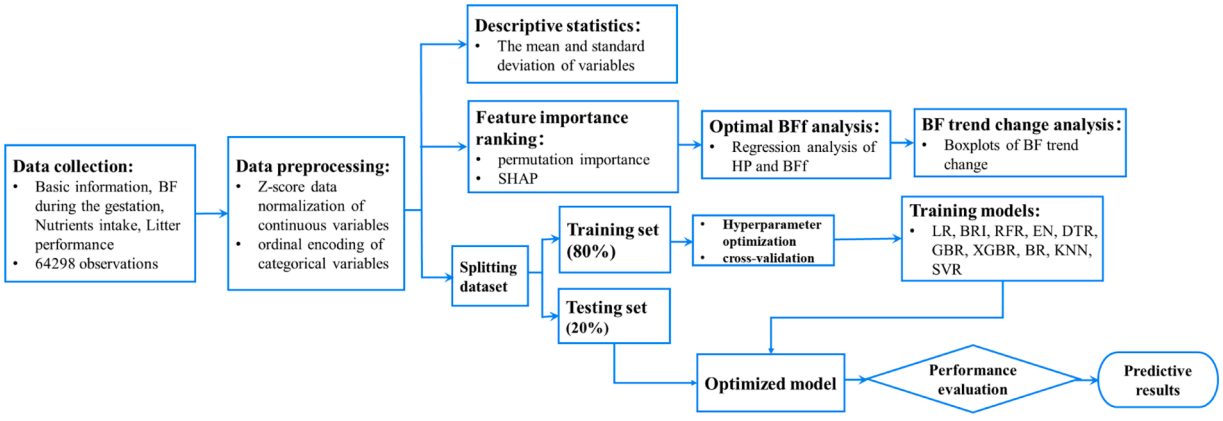


Fig. 1. Flowchart of the study for analysis and prediction of sow BF.

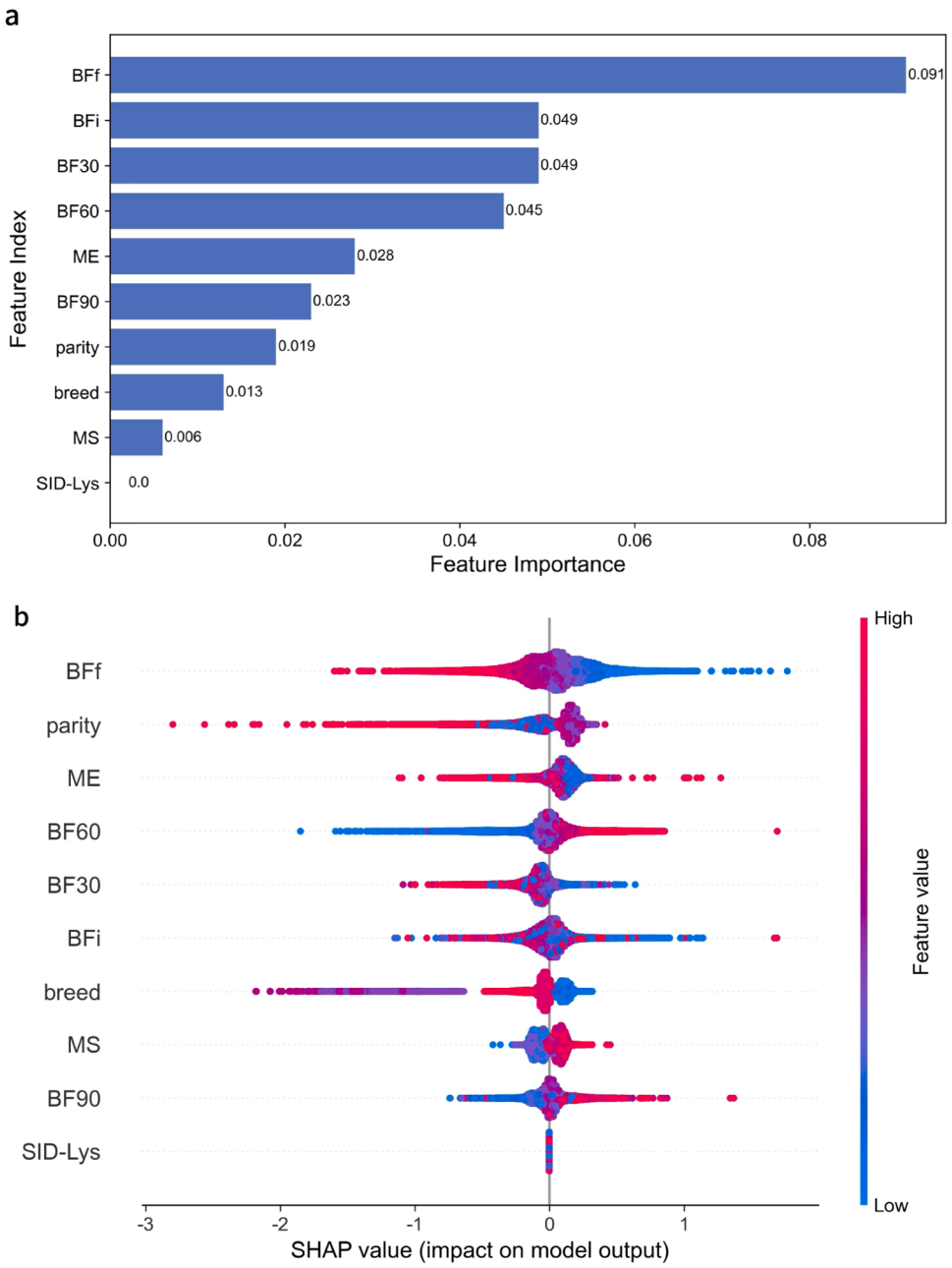


Fig. 2. Ranking of the feature importance. a: permutation importance in the XGboost model; b: the features were ranked by the sum of SHAP values of all observations. The SHAP values were used to show the distribution of each feature. Red indicated high values and blue indicated low values of the feature. The x-axis represented the influence of the SHAP value on the model output.

requires more computational resources and time to run, and also increases the difficulty of model tuning by introducing new hyperparameters. Therefore, considering the basic performance of the model and computational costs in this study, we chose to adjust the hyperparameters without combining with meta-heuristic algorithms. We tried using Particle Swarm Optimization (PSO) to optimize hyperparameters, but its performance wasn't as good as that of models optimized using grid search.

The Grid Search method was used to systematically tune the hyperparameters. For each model, core hyperparameters were selected that significantly affect the model performance. The range of candidate values was set according to domain knowledge or literature references, and constructed a discretized hyperparameter space. Table 1 showed the hyperparameters of each model. The data were randomly divided into two parts: 80 % for training set and 20 % for testing set. For each hyperparameter combination, a 5-fold cross-validation scheme was employed for training.

2.4. Evaluation of model performance

To evaluate the performace of different models in predicting BF at different gestation period, this study adopted 3 metrics:the coefficient of determination (R^2) (Eq. (2)), root mean square error (RMSE) (Eq. (3)), and mean absolute percentage error (MAPE) (Eq. (4)). Higher R^2 indicated better model fitting, while lower RMSE and MAPE indicated better model performance. These three metrics are defined as follows:

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (\bar{y} - y_i)^2} \quad (2)$$

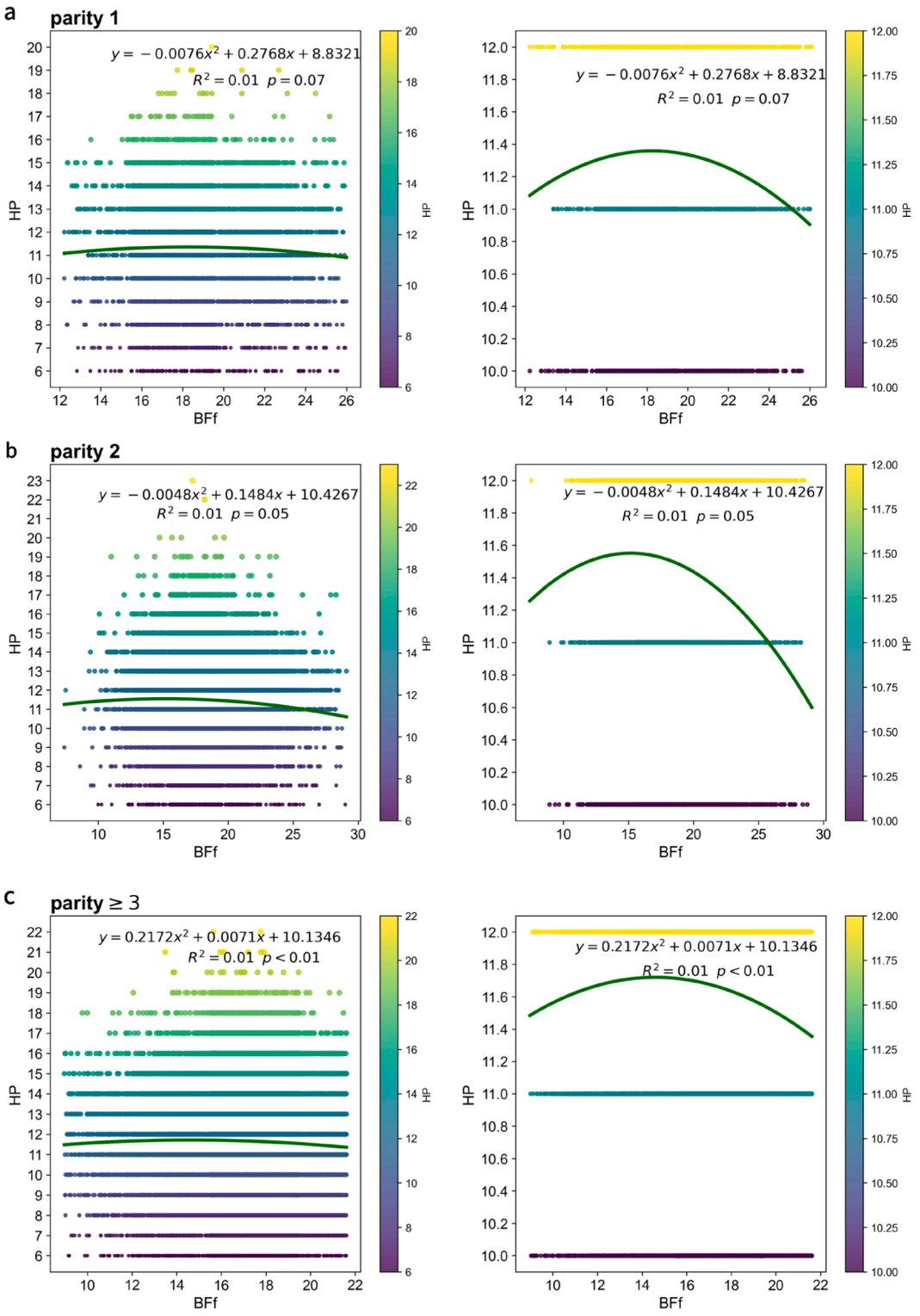


Fig. 3. The optimal BFf for sows of different parities. a∼c were scatter plots and regression curves between BFf and HP of the parity 1, parity 2, and ≥3, respectively.

$$RMSE = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(\hat{y}_i - y_i)^2} \tag{3}$$

$$MAPE = \frac{1}{m}\sum_{i=1}^{m}\left|\frac{\hat{y}_i - y_i}{y_i}\right| \times 100 \tag{4}$$

where m represents the total number of observations, $\hat{y}_i$ represents the ith predicted value, y_i represents the ith observed value, $\bar{y}$ represents the average of the observed values.

An overall flowchart of the study is provided in Fig. 1.

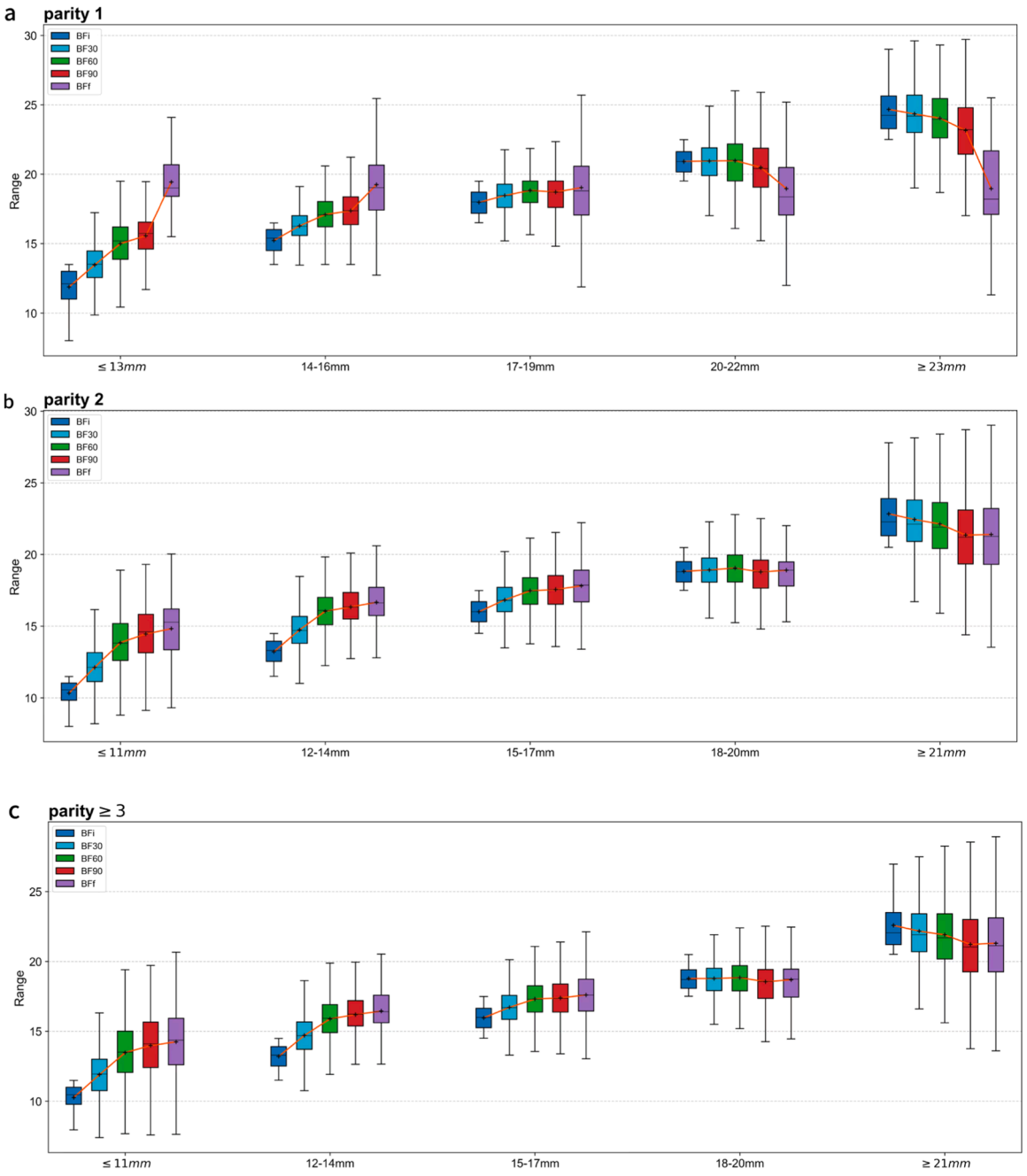


Fig. 4. Boxplots of BF trend change during gestation for sows of parity 1 (a), parity 2 (b), and ≥3 (c) with different BFi. Black line in the middle of the boxplot represents the median, and the cross line represents the mean.

Table 3
Performance comparison of 10 models for predicting BF30, BF60, BF90 and BFf.

Model	BF30			BF60			BF90			BFf		
	R^2	RMSE, mm	MAPE, %	R^2	RMSE, mm	MAPE, %	R^2	RMSE, mm	MAPE, %	R^2	RMSE, mm	MAPE, %
LR	0.85	1.19	5.67	0.83	1.11	5.03	0.84	1.03	4.55	0.90	0.82	3.60
BRI	0.85	1.19	5.67	0.83	1.11	5.03	0.84	1.03	4.55	0.90	0.82	3.60
RFR	0.85	1.18	5.61	0.84	1.10	4.96	0.84	1.02	4.49	0.90	0.81	3.57
EN	0.84	1.22	5.80	0.82	1.15	5.19	0.83	1.06	4.71	0.89	0.85	3.74
DTR	0.84	1.20	5.70	0.83	1.12	5.05	0.84	1.03	4.55	0.90	0.82	3.61
GBR	0.85	1.17	5.58	0.84	1.09	4.94	0.84	1.01	4.44	0.90	0.81	3.54
XGBR	0.85	1.17	5.57	0.84	1.09	4.93	0.84	1.01	4.46	0.90	0.81	3.55
BR	0.84	1.20	5.73	0.83	1.12	5.06	0.83	1.04	4.61	0.90	0.83	3.66
KNN	0.72	1.61	7.45	0.71	1.46	6.57	0.69	1.42	6.28	0.73	1.34	5.70
SVR	0.51	2.12	9.63	0.48	1.96	8.57	0.47	1.86	7.97	0.59	1.66	6.72

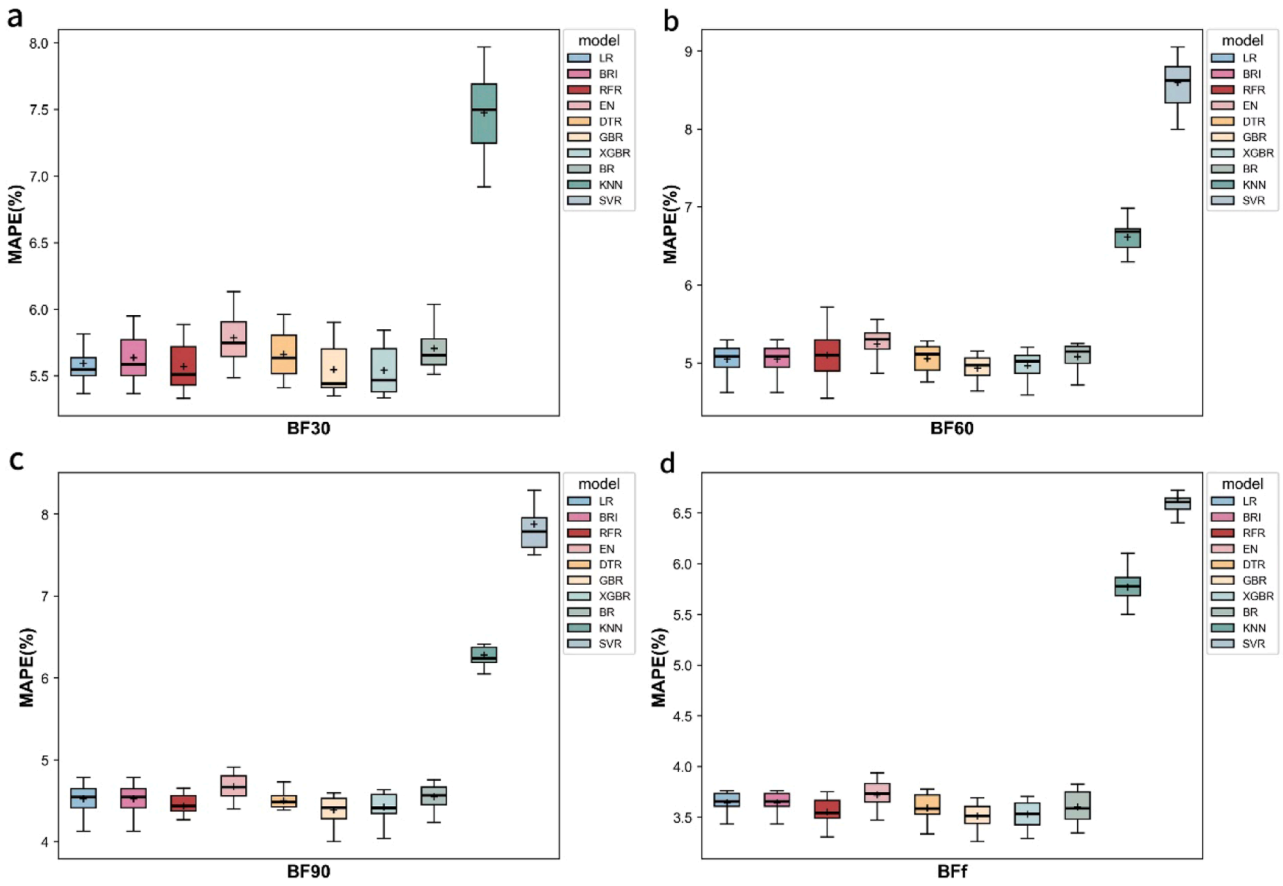


Fig. 5. MAPE boxplot of 10 models. In the boxplot, the central black line represents the median, the cross line represents the mean and the cross line represents the mean.

3. Results

3.1. Preliminary statistical analysis of the data

Table 2 presented a statistical analysis of the mean and standard deviation (SD) of parity, mating age (MA), BF, nutrient intake and litter performance of the sows. Overall, the average BF increased 1.53 mm from insemination to farrowing, and during the early 30 days the BF increased 0.71 mm, higher than other gestation stages. Correspondingly, the average daily nutrient intake during the early 30 days was higher than middle gestation, but lower than late gestation. This indicated that nutrient intake in the early stage of gestation were more conducive to the maternal recovery and growth.

3.2. Feature importance ranking for HP

The result of permutation importance demonstrated that BFf was more important feature for HP, indicating that the BFf was key factor affecting HP of sows, and even more important than BF of other gestation stages (Fig. 2a). Furthermore, the result of SHAP method to evaluate magnitude of effect on the HP presented that excessive BFf reduced the number of HP. Regarding the parity, middle parities had a positive effect, while extremely high or low parities exhibited a negative effect on HP (Fig. 2b).

3.3. Determination the optimal BFf of different parities and the trend change of BF during gestation

Based on the importance of BFf on HP of sows, we conducted a regression analysis on the optimal BFf of different parities. The results showed that the optimal BFf of parity 1, 2, and ≥3 was 18, 16, and 16 mm, respectively (Fig. 3a~3c).

The BFi of all observations were divided into 5 grades according to the optimal BFf of different parities. The data was divided into ≤13 mm, 14~16 mm, 17~19 mm, 20~22 mm, ≥23 mm in parity 1 and ≤11 mm, 12~14 mm, 15~17 mm, 18~20 mm, ≥21 mm in parity 2 and parity ≥3. The trend change of BF during gestation different parities and BFi was assessed (Fig. 4a~4c). The low BFi of sows presented an upward trend during gestation in all parities, the lower BFi showed the greater increase in the gestation. On the contrary, high BFi of sows presented a downward trend during gestation in all parities, the higher BFi showed the greater decrease in the gestation. In addition, low BFi showed higher increase of BF in the early and middle stages of gestation than late gestation in multiparous sows. The similar trend change was also observed for high BFi of sows with a decrease. This indicated that the early and middle gestation of gestation were the best periods for body condition recovery or BF adjustment.

3.4. Comparison of different models performance for different BF during gestation

Table 3 presents the R^2, RMSE and MAPE of 10 models for the prediction of BF30, BF60, BF90, BFf. The R^2, RMSE, and MAPE were 0.55~0.85, 1.17~2.12 mm and 5.57 %~9.63 % of BF30, respectively; The R^2, RMSE and MAPE were 0.48~0.84, 1.09~1.96 mm and 4.93 %~8.57 % of BF60, respectively; The R^2, RMSE and MAPE were 0.47~0.84, 1.01~1.86 mm and 4.44 %~7.97 % of BF90, respectively; The R^2, RMSE and MAPE were R^2, RMSE, and MAPE 0.59~0.90, 0.81~1.66 mm, and 3.54 %~6.72 % of BFf, respectively. For the prediction of BF30 and BF60, XGBR displayed the lowest RMSE and MAPE

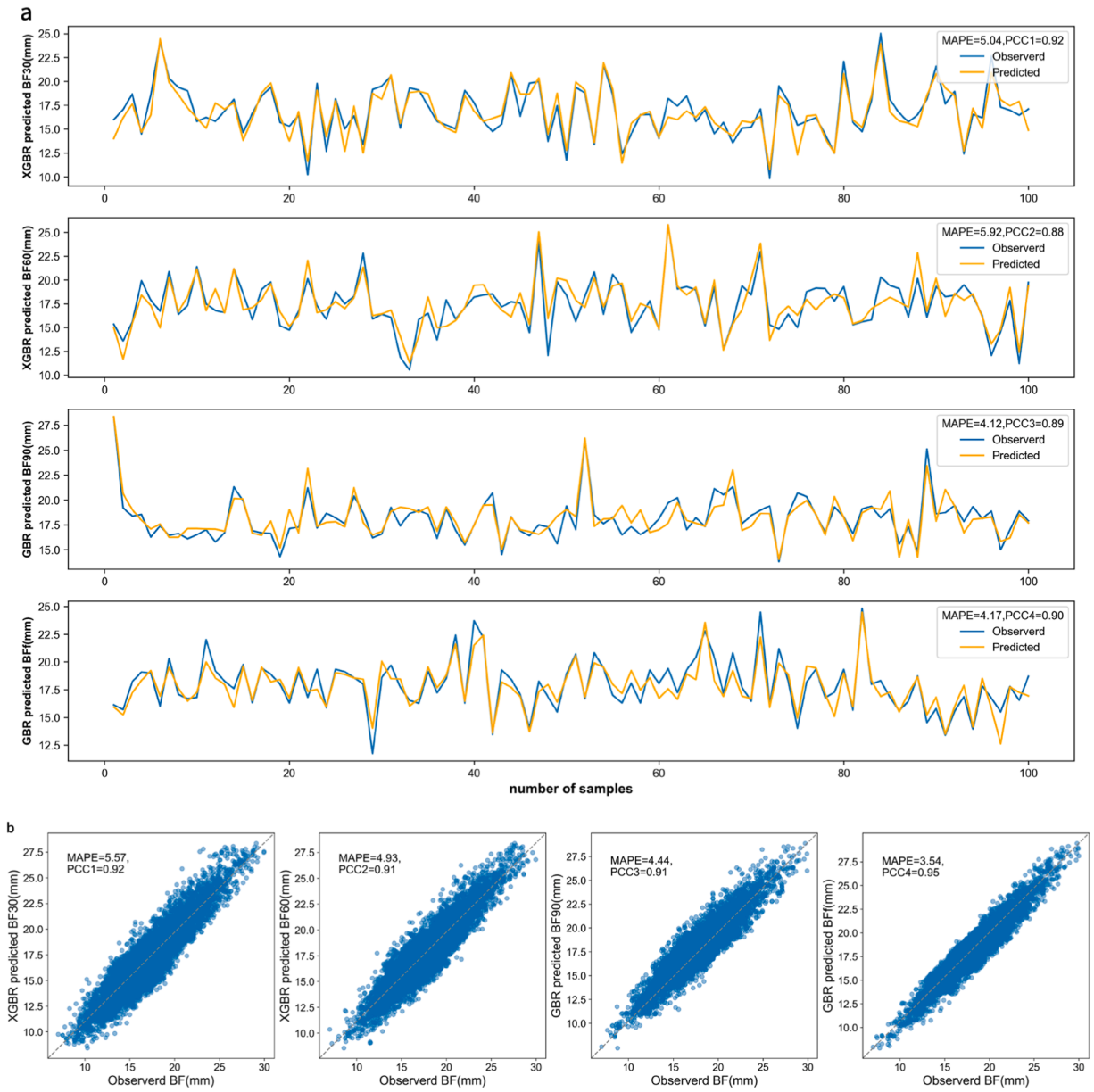


Fig. 6. The predicted and observed values for XGBR and GBR models. a: observed and predicted data values using different models. b: scatter plot of predicted and observed values.

compared to other models (RMSE were 1.17 mm and 1.09 mm of BF30 and BF60, respectively. MAPE were 5.57 % and 4.93 % of BF30 and BF60, respectively). For the prediction of BF90 and BFf, GBR exhibited the lowest RMSE and MAPE compared to other models (RMSE were 1.01 mm and 0.81 mm of BF90 and BFf, respectively. MAPE were 4.44 % and 3.54 % of BF90 and BFf, respectively). The results indicated the out-performance of two models in predicting BF. Furthermore, the MAPE boxplots were used to measure the stability of different models (Fig. 5a~5d). The median of MAPE about XGBR and GBR was lower than other models, with smaller fluctuation and dispersion. Therefore, XGBR and GBR were the best models for predicting BF of gestating sows.

A total of 100 points were randomly selected from the testing dataset to compare the predicted values and observed values by XGBR and GBR models (Fig. 6a). The MAPE and correlation coefficients between observed and predicted values for XGBR, and GBR models were 5.04 %, 5.92 %, 4.12 %, and 4.17 %, and 0.92, 0.88, 0.89 and 0.90, respectively. In addition, the scatter plots predicted and observed by the XGBR and GBR models were closer to the 1:1 line (Fig. 6b), indicating that the XGBR and GBR models had excellent performance in predicting BF.

4. Discussion

4.1. Feature importance ranking for reproductive performance of sows

The reproductive performance of sows is influenced by numerous factors such as breed, parity, environment, nutritional management, and body condition [1–3]. It is well known that breed has a significant impact on litter performance of sows. Compared to purebred sows, crossbred sows display higher number of TB and birth weight [10,33, 34]. With the increase in parity, litter performance increases in middle parities and decreases in high parities. The TB of sows of parity 1 and ≥6 are lower than parity 2~5 [1,35,36]. In previous studies, high temperature in summer can reduce the TB and conception rate of sows due to the high temperature decreases the secretion of gonadotropin-releasing hormone and damages the development of ovarian follicles, resulting to the injury of corpus luteum and decrease in progesterone concentration

Table 4
Comparison of BF prediction in different studies.

Model	Dataset	Evaluation and results of model performance	References
Linear regression	731 sows of BF and body condition scores	R^2 0.19	[28]
Multiple linear regression based on machine vision	BF and external morphological parameters of 100 sows	R^2 0.8923 MAE 1.23mm MAPE 5.73 %	[29]
Deep neural networks	3782 head-side images and each image contains two labels of BF and segmentation mask of BF area	R^2 0.915 MAE 1.275mm MAPE 6.4 %	[52]
Multiple linear regression based on machine vision	3481 longitudinal images from 574 pigs	R^2 0.58 RMSE 1.65mm	[53]
Residual Network-Based Model	Video data of 48 sows	R^2 0.89 MAE 0.66mm	[22]
XGBR and GBR	64,298 observations of sow basic information, BF, nutrients intake, litter performance	R2 0.84∼0.90 RMSE 0.81∼1.17mm MAPE 3.54 % ∼5.57 %	Our study

[35,37–40]. Feeding management during gestation regulates the body condition of sows through nutrient allocation and supply, with appropriate BFf resulting in superior reproductive performance such as birth weight, vitality, and weaning weight of piglets [2–6].

In this study, permutation feature importance and SHAP methods proved that BFf showed greatest contribution to HP compared to other features, is the most important feature affect HP. Lower BFf represented positive effect on HP, and excessive BFf had a negative effect through the SHAP value, which was consistent with the previous studies [6,13,18]. Parity and ME intake were second important features. High parity and low parity were not conducive to HP, consistent with previous studies. However, excessive ME intake negative influence on HP perhaps was related to the feeding program, which the sows with low BFi were supplied high feed intake, but didn't completely reach the optimal BFf.

4.2. Determining the optimal BFf and the trend change of BF during gestation

On the basis of feature importance ranking for HP, we used scatter plots and linear regression analysis to determine the ideal BFf of different parities. The relationship between reproductive performance and BFf was commonly uesd to determine the optimal BFf, and the recommended ideal BFf was 14∼22 mm in previous studies [6,13,17–20]. The recommended optimal farrowing backfat range varies probably related to factors such as the genotype, farms (feeding management, environmental conditions) [41]. In recent years, the recommended optimal BFf has declined might be the decrease of body fat and backfat deposition. In our study, >60,000 observed values were demonstrated the optimal BFf of parities 1, 2, and ≥3 were 18 mm, 16 mm, and 16 mm, respectively. Compared with other studies, the dataset was larger and more representative of the population. Therefore, it is feasible and practical to determine the optimal BFf based on the big data analysis in breeding farms.

The trend changes of BF from insemination to farrowing indicated the recovery pattern of body condition during gestation. Sows with lower BFi were supplied with higher feed intake displayed the larger increase in BF during gestation, in accordance with the results that early and middle stages were the ideal periods for body condition recovery and backfat adjustment in previous studies [42–44]. Therefore, it is necessary to focus on the feed intake in the early and middle stages of gestation to formulate feeding program based on the BF of gestating sows.

4.3. Effects of different ML models on the prediction of BF

The comparison of related studies to BF prediction was shown in Table 4. Compared with other studies, our study displayed a large sample size of dataset over than 60,000 observations result in the models were more reliable. Moreover, regarding the estimation of BF in our study, other gestation periods of BF could be able to make predictions in advance, which is beneficial to evaluate the nutrient requirements or adjust the feeding plan timely. This study employed 10 ML models to predict the BF30, BF60, BF90 and BFf. LR is a simple model widely used to predict and calculate the overall trend, but it couldn't be accurately predicted and applied under the influence of nonlinear factors [45]. In contrast, the other 9 models performed better understand the nonlinear relationship between various factors and BF. Among them, XGBR and GBR showed excellent prediction performance with the results of comparison chart of predicted values and measured values (Fig. 6a) and the scatter plot (Fig. 6b). GBR model can improve the fitting effect by training multiple learning residual tree models, and it can directly handle discrete features by converting them into continuous features [46]. XGBR can handle sparsity and avoid overfitting through shrinkage and regularization techniques based on gradient tree boosting [47,48]. XGBoost is specifically designed to optimize memory usage and utilize the computing power of the hardware, reduced the computing time and improving the performance compared with other ML models [49]. The XGBR and GBR models had achieved good predictive performance in other studies. Durand et al. [50] used the GBR model to predict the energy and SID-Lys of gestating sows, with R^2 of 0.95 and 0.99, respectively, and MAEP of 1.55 % and 0.82 %, respectively. Gertz et al. [51] employed the XGBoost model to evaluate sensor data and health records to distinguish between healthy and sick dairy cows. Ultimately, the performance attained area under the curve of 86 % (±2 %), F-value of 81 % (±2 %) and a relatively balanced specificity (78 %) and sensitivity (81 %) level, indicating that XGBoost was a convenient and effective method for identifying disease behaviors.

5. Conclusion

This study applied over 60,000 observations about BF and production performance to determine that BFf was the most important feature affecting the HP of sows. The optimal BFf for sows of parity 1, 2, and ≥3 were 18 mm, 16 mm, and 16 mm, respectively. In addition, it was clear that the early and middle stages of gestation were identified as the best periods for body condition recovery through the trend changes of BF during the gestation. Furthermore, we established that the XGBR and GBR models could promptly and accurately predict the BF of sows during gestation with superior prediction accuracy and stability compared with other models. These findings contribute to understanding the maternal recovery and body condition reserve status of gestating sows, accurately predicting the nutrient requirements, timely and effectively intervening in the feeding management of sows in advance, providing a solid foundation for implementing and formulating precise feeding, improving the reproductive performance and production benefits.

Ethics statement

Animal use approval was not required because no animals was harmed in this study, the data were obtained from production recording system of Guangxi State Farms Yongxin Livestock Husbandry Group Co., Ltd.

CRediT authorship contribution statement

Xuewu Peng: Writing – original draft, Methodology, Investigation, Conceptualization. **Yaxin Song:** Software, Formal analysis. **Yuanfei Zhou:** Methodology, Conceptualization. **Hongkui Wei:** Validation, Resources. **Siwen Jiang:** Supervision, Project administration. **Fukang Wei:** Resources, Data curation. **Xinran Li:** Writing – review & editing, Visualization, Methodology, Formal analysis. **Jian Peng:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request. Data will be made available on reasonable request. A request for accessing data may be sent to Xinran Li(xinran.li@mail.hzau.edu.cn).

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